TEXORA · SESSIONAL NOTES · STUDY SHEET

# Dry Wash Processes of Garments

*The waterless and low-water effect processes of the garment laundry — hand sanding, whiskering, permanganate spray, resin curing, laser fading, ozone treatment and dry ice blasting — studied as controlled trials on denim panels, with safety and repeatability as the measures that matter.*

AM-4102 · ADVANCED APPAREL MANUFACTURING-I SESSIONAL · STUDY 2 · LEVEL 4 TERM 1

## 1. Aim and place of the study

**Aim:** to perform, observe and compare the dry processes used to create localised fashion effects on garments — chiefly denim — recording for each the mechanism, the control variables, the effect quality and the operator-safety provisions, and to judge when a dry process replaces a wet one.

"Dry wash" in the garment laundry does not mean dry cleaning. It is the family of **effect processes applied to the dry garment**, before (or instead of) the wet wash of Study 1: they create the *pattern* of wear — faded thighs, whiskers, sanded edges — which the subsequent wet wash then softens and blends. The division of labour is fundamental: **dry processes localise, wet processes generalise.** A modern vintage jean is a sequence: dry effects first, then enzyme/stone/bleach wash, then tinting and softening.

## 2. Materials, machines and specimens

**Equipment:** inflatable mannequin/forms and sanding station with dust extraction; emery paper grades 320–600 and sanding blocks; whisker templates and hot-air/press equipment for resin setting; spray booth with extraction and full PPE (respirator, goggles, gloves) for permanganate spray; laser fading machine (demonstration/video study where the institute machine is unavailable); ozone chamber or ozone-generator cabinet (demonstration); scraping tools, brushes and tag guns.

**Specimens:** rigid denim jeans (same lot as Study 1) and flat denim panels 40 × 40 cm mounted on boards — panels for sanding, spray and laser trials so variables can be changed one at a time; one full garment for the combined whisker + sanding trial.

## 3. The dry process family

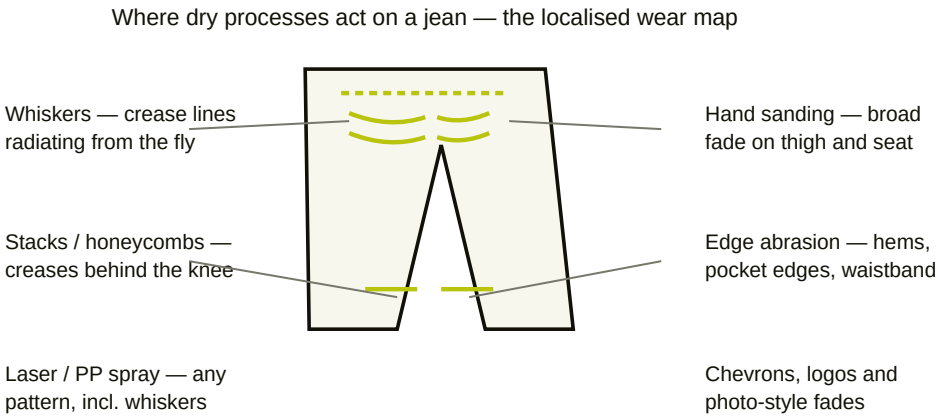



Figure 1 — The wear map of a vintage jean. Dry processes reproduce the places a body actually abrades and creases denim.

| PROCESS | MECHANISM | EFFECT AND TYPICAL USE | CONTROL VARIABLES / HAZARDS |
|---|---|---|---|
| Hand sanding | Emery paper rubbed by hand over the garment on an inflatable form removes indigo-bearing surface fibres. | Broad soft fading of thighs, seat, knees; the base effect of most vintage styles. | Paper grade, pressure, strokes count. Dust extraction essential; operator RSI and dust exposure. |
| Whiskering (creasing) | Creases are formed by hand or template, fixed with resin (or heat) and cured; the raised crease lines then abrade lighter in sanding/washing. | Radiating crease lines at the front hip ("cat whiskers"), honeycombs behind knees. | Template symmetry, resin dose, cure temperature/time. Formaldehyde-based resins need controlled handling. |
| PP spray | Potassium permanganate solution sprayed on selected zones oxidises indigo locally; neutralised in the following wash. | Bright local discharge, chevrons, logos, intensifying sanded areas. | Spray concentration, distance, air pressure. Respirator and booth extraction compulsory; $MnO_2$ staining until neutralised. |
| Laser fading | A computer-guided $CO_2$ laser vaporises surface dye/fibre along a programmed image — no contact, no chemicals. | Any artwork: whiskers, patterns, logos, photo effects; perfectly repeatable. | Power, speed, resolution (dpi), passes. Excess power cuts/holes the fabric; eye protection and fume extraction. |

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|---|---|---|---|
| Ozone treatment | Ozone gas ($O_3$) in a sealed chamber oxidises indigo over the whole garment or, with resists, in zones. | Overall back-side lightening, bleach substitute, "used" tone; also deodorises. | Gas concentration, humidity, time. Ozone is toxic — sealed chamber, destruct unit, leak monitoring. |
| Dry ice / other mechanical | Blasting with $CO_2$ pellets or air/abrasive jets erodes the surface without water. | Soft surface frosting, low-water alternative for some effects. | Nozzle pressure and distance; cold burns, noise, cost per garment. |

## 4. Procedure

1. **Sanding trial (panels):** sand three panels with grades 320, 400 and 600, 40 strokes each at consistent pressure on the form. Record which grade reaches the target fade (compare with the Study 1 washed specimens) with least visible fibre damage under the hand lens.
2. **Whisker trial (garment):** mount the jean on the form, form front whiskers by hand to the template, pin, apply resin by brush to the crease zone, cure per the resin data sheet (record temperature and time), cool, then sand lightly across the creases.
3. **PP spray trial (panel, booth):** mask a chevron; spray permanganate solution at the recorded concentration and distance in two passes; note the brown $MnO_2$ colour — the true discharge appears only after neutralising in a metabisulphite bath (link to Study 1, specimen E).
4. **Laser study:** from the machine demonstration or program file, record the settings used for a whisker pattern (power %, speed, dpi) and inspect the result for edge sharpness and any fibre damage at high power.
5. **Ozone study:** record chamber conditions (concentration, humidity, time) for a demonstration load and compare the shade drop achieved with the bleach-wash result of Study 1 at similar input cost.
6. **Evaluation:** assemble all trials; grade each effect for (a) match to the target standard, (b) naturalness (does it sit where a body wears?), (c) fabric damage, (d) repeatability garment-to-garment, (e) operator exposure controls in place.

## Observation points — what to record

- For sanding: grade vs strokes vs fade grade, and the point at which fibre damage (hairiness, thinning) becomes visible.
- Whisker geometry: symmetry left/right, crease count and spacing against the template.
- PP spray: concentration, distance, passes — and the shade after neutralising, not before.
- Laser: the power threshold at which fading becomes cutting (from the demonstration samples).

- A comparison table: minutes of labour per garment and water use per garment for dry vs wet routes to the same target look.

## Precautions

- Dust: sanding without extraction exposes the operator to cotton/indigo dust — the station extraction runs before the paper touches cloth, and masks are worn.
- PP spray only in the booth with respirator; permanganate is a strong oxidiser — keep from acids and organic stores.
- Laser and ozone equipment are demonstration-only for students: interlocks, sealed chambers and destruct units are never bypassed.
- Resin chemicals: gloves; curing ovens are burn hazards; record cure conditions — under-cured resin washes out and the whiskers vanish in the first wash.

## Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. State the division of labour between dry and wet garment processes, with one example of each.
2. Why are whiskers resin-cured *before* sanding and washing rather than created by the wash itself?
3. Give two advantages and two limitations of laser fading against hand sanding.
4. Why is the shade of a PP-sprayed panel judged only after neutralising?
5. An ozone chamber lightens a whole load evenly. Why can it not, by itself, produce whiskers?

### Answer key

1. Dry processes localise effects to the wear map (sanding a thigh fade, spraying a chevron); wet processes generalise — softening, overall fading, cleaning and blending the dry effects into a natural whole (enzyme wash after sanding).
2. The wash is a general bath — it cannot create sharp lines. The crease pattern must be physically set into the garment first (resin/heat); abrasion and washing then lighten the raised crease ridges, making the set lines visible as whiskers.
3. Advantages: perfect repeatability of an artwork file and no chemical/water use at the point of effect (also much faster per garment at volume). Limitations: high capital cost; excess power burns or cuts the cloth, and the flat, printed look can appear unnatural next to hand work if the artwork is poor.
4. Fresh PP spray shows brown $MnO_2$, which masks the true result. Neutralising (metabisulphite) removes the manganese stain and reveals the actual discharged shade — judging before that step judges the wrong colour.

5. Ozone is a gas acting on every exposed surface of the load; it has no spatial pattern. Whiskers are a pattern — they need a localising step (creasing, masking/resist or laser) to define where the effect sits.

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